

Benchmarking Vision-Language Models for Robotic Perception and Planning

From Object Grounding to Hierarchical Task Execution

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Abstract

We present a five-stage benchmark evaluating Vision-Language Models (VLMs) for robot perception, grasp reasoning, and sequential task planning. Five state-of-the-art VLMs—GPT-5.4 Mini, Gemini 3.1 Flash Image, Qwen 3.5 9B, Seed 2.0 Lite, and Ministral 8B—are evaluated through a progressive set of robotic manipulation tasks requiring structured JSON outputs.

Test 1 evaluates isolated single-object grasping. **Test 2** introduces multi-object scenes requiring target localization. **Test 3** increases scene clutter and enforces strict spatial grounding. **Test 4** evaluates grounded multi-step execution planning in real-world domestic scenes. **Test 5** evaluates high-level task decomposition and autonomous planning under varying levels of task ambiguity.

Tests 1–3 use a $[0, 5]$ scoring scale and show that GPT-5.4 Mini consistently achieves the strongest perception and grasping performance ($\mu = 4.88, 4.80, 4.73$), while Seed 2.0 Lite demonstrates remarkable stability across increasing perceptual difficulty. Open-weight models, particularly Ministral 8B, exhibit severe degradation under strict localization constraints.

In Test 4 ($[0, 1]$ scale), Gemini Flash Image achieves the highest execution planning score ($\mu = 0.751$), narrowly surpassing GPT-5.4 Mini ($\mu = 0.745$), while Qwen 3.5 9B shows catastrophic variance ($\sigma = 0.391$). Test 5 confirms the dominance of GPT-5.4 Mini ($\mu = 0.788$) and Gemini Flash Image ($\mu = 0.785$) for hierarchical task decomposition and grounded planning. Across all planning-oriented evaluations, Scene Understanding emerges as the primary bottleneck criterion.

We further introduce a **grasp taxonomy analysis** examining whether the five VLMs share consistent definitions of standard grasp types. We find strong consensus on simple grasps (power, hook, cylindrical) and significant divergence on boundary cases (pinch, pincer, palmar, lateral), revealing that VLMs encode different operational definitions of the same physical primitives—a critical concern for robot systems relying on VLM-generated grasp plans.

Inter-evaluator agreement in Test 1, recomputed after three independent runs per LLM judge, rises from $r = 0.50$ (single run) to $r = 0.91$ (3-run averaged), and reaches $r = 0.931$ in the structured high-level planning task (Test 5), confirming that both *rubric structure* and *multi-run averaging* are required for reliable LLM-as-judge benchmarks.

1 Introduction

This paper presents a five-test benchmark that progressively increases task complexity along two axes: *perceptual difficulty* (Tests 1–3) and *planning depth* (Tests 4–5). [1, 2]

Test 1 evaluates isolated single-object grasping. Test 2 introduces multi-object scenes requiring target localization. Test 3 adds clutter and enforces a stricter spatial grounding rubric. Test 4 shifts from perception to grounded *execution planning*, where the model must generate a complete sequence of physically coherent manipulation steps for a real-world scene. Finally, Test 5 evaluates *high-level autonomous planning*: the model must decompose an underspecified task into atomic subtasks before producing a grounded execution strategy.

Beyond evaluating task performance, we investigate a subtler issue: do the five VLMs even agree on what a *grasp* is? We find that models share a common framework (Napier [12], Cutkosky [13], Feix [14]) but apply it with markedly different levels of rigour, diverging precisely on the boundary cases most relevant to dexterous manipulation.

The remainder of this paper is structured as follows. Section 2 reviews relevant work. Section 3 describes the evaluation protocols. Section 4 reports Test 1 (single object). Section 5 reports Test 2 (multi-object). Section 6 reports Test 3 (strict localization). Section 7 analyses inter-model grasp taxonomy consistency. Section 8 reports Test 4 (multi-step execution planning). Section 9 reports Test 5 (high-level autonomous planning). Section 10 provides a cross-test synthesis. Section 11 discusses implications and limitations.

1.1 Contributions

This work makes five primary contributions:

1. We introduce a progressive five-stage benchmark spanning object perception, localization, grasp reasoning, execution planning, and high-level task decomposition.
2. We propose structured scoring protocols for both low-level grasping and high-level robotic planning tasks using dual LLM evaluators, and demonstrate that multi-run averaging is essential for reliable agreement.
3. We provide one of the first comparative analyses of proprietary and open-weight VLMs across increasingly complex robotics-oriented reasoning tasks.
4. We identify scene understanding and spatial grounding as the primary bottlenecks limiting current VLM deployment in robotic manipulation systems.
5. We introduce a *grasp taxonomy consistency analysis*, revealing that VLMs diverge on the definition of four standard grasp types in ways that would produce contradictory robot motion plans for the same object.

2 Related Work

Existing VLM benchmarks address either (i) general visual question answering [3, 4, 5], (ii) bounding-box grounding [6], or (iii) embodied policies inside specific robot stacks [7, 8, 9, 10]. None of them jointly evaluate zero-shot grasp reasoning, textual spatial localization without coordinates, full execution planning, and autonomous task decomposition under linguistic ambiguity, using a unified scoring protocol and dual LLM judges. Our benchmark closes exactly this gap and is, to our knowledge, the first to chart the *capability degradation curve* of frontier VLMs as a function of robotics-relevant task complexity.

VLM benchmarks. Standard multi-modal benchmarks such as VQA [3], MMMU [4], and MMBench [5] focus on global visual understanding and question answering. These benchmarks do not assess robot-centric affordance reasoning or structured grasp planning, motivating our task-specific evaluation.

Object localization in VLMs. Grounding-capable models such as Grounding DINO [6] address object localization through bounding-box prediction. Our benchmark evaluates a complementary skill: *textual* localization of an object within a scene as part of a downstream manipulation plan, without requiring spatial coordinates.

Grasp taxonomies. Foundational grasp classifications by Napier [12] (power vs. precision), Cutkosky [13] (16-type taxonomy), and Feix et al. [14] (GRASP taxonomy, 33 types) remain the standard references. We use these as the ground truth against which VLM definitions are compared.

Robotic grasping and language. Language-conditioned manipulation has been studied in RT-1 [7], RT-2 [8], and SayCan [9]. PaLM-E [10] and similar embodied models demonstrate VLM integration in closed-loop robotic systems. Our work provides a *task-agnostic* quantitative evaluation of raw VLM capabilities before any robotics fine-tuning.

LLM-as-judge evaluation. We use two LLM judges to score VLM responses, following the LLM-as-judge paradigm [11]. Inter-evaluator agreement is measured via Pearson r , Spearman ρ , and MAE across both evaluators, allowing us to quantify scoring reliability as a function of task structure.

3 Evaluation Protocol

3.1 Models

The five VLMs evaluated across all tests are described in Table 1.

Table 1: VLMs evaluated. “—” = undisclosed parameter count.

Model	Type	Params
GPT-5.4 Mini	Proprietary	—
Gemini 3.1 Flash Image	Proprietary	—
Seed 2.0 Lite	Proprietary	—
Qwen 3.5 9B	Open-weight	9B
Ministral 8B	Open-weight	8B

3.2 Object Dataset

Eight everyday objects are used across Tests 1–3: *blue pen*, *grey mug*, *computer mouse*, *orange balloon*, *blue can*, *USB drive*, *whiteboard eraser*, and *stirrer*. In Test 1 each object is presented in isolation; in Test 2 the same object appears in a multi-object scene alongside visual distractors. In Test 3, the same objects appear in a cluttered scene where some are partially occluded. Tests 4 and 5 use 10 photographs of real domestic environments (table after a meal, kitchen, bathroom, laundry room, sink, etc.).

3.3 Scoring System — Tests 1–3

These tests use an identical five-criterion rubric. Each criterion is scored 0–5; the final score is their arithmetic mean.

(A) Identification. *Test 1*: name match (exact=3, synonym=1.5, wrong=0) plus attribute bonus (+0.5 per valid property, max 2). If wrong name: cap at 2. *Tests 2–3*: the name sub-score is replaced by a *position score* (+1 per valid spatial descriptor, max 3); attribute bonus unchanged. Describing the wrong object yields A=0.

(B) **Planning (0–5)**. Approach 0.5 pt, pre-grasp 0.75 pt, grasp 1.0 pt, lift 0.75 pt (= 3 pts actions); logical order 1 pt + efficiency 1 pt (= 2 pts quality).

(C) **Use (0–5)**. Realism (2 valid uses=2, 1=1), grasp-use link (explicit=2, implicit=1, absent=0), physics plausibility 1 pt.

(D) **Grasp (0–5)**. Type correctness (2 pts), contact zone precision (2 pts), accessibility 1 pt.

(E) **Robustness (0–5, rescaled)**. Hallucination (none=1, minor=0.5, major=0) + logic (perfect=1, minor=0.5, incoherent=0). Raw score $\in [0, 2]$ rescaled: $E_{\text{scaled}} = E_{\text{raw}} \times 2.5$.

Final score. Score = $(A + B + C + D + E_{\text{scaled}})/5$.

3.4 Dual-Evaluator Validation

Each VLM response is independently scored by **GPT-5.2** and **Claude Sonnet 4.6** acting as LLM judges [11]. All reported scores are means over both evaluators. Inter-evaluator agreement is measured by Pearson r , Spearman ρ , and Mean Absolute Error (MAE). In Test 1, three independent runs per evaluator are averaged before correlating, following the hierarchical averaging procedure described in Section 4.

4 Test 1: Single-Object Grasping

4.1 Task

The VLM receives a single image of one object with no scene context and produces a JSON response with five sections: `object_identification` (name + attributes), `grasp`, `planning`, `use`, `robustness`.

4.2 Multi-Run Averaging Protocol

To address the low inter-evaluator agreement observed in the original single-run evaluation ($r = 0.50$), two additional independent runs were collected per LLM judge, yielding 3 runs $\times$ 2 evaluators $\times$ 5 models $\times$ 8 objects = 240 rated responses. All Test 1 scores are computed by first averaging each evaluator’s 3 runs at the (model, object) level, then averaging the two evaluators—preserving the original dual-evaluator protocol while substantially reducing per-judge noise before correlation.

4.3 Results

Table 2 reports updated subscores with 95% bootstrap confidence intervals (10 000 resamples over 8 objects) and within-evaluator run standard deviation $\bar{\sigma}_{\text{run}}$. GPT-5.4

Mini leads with $\mu = 4.88$ and perfect Planning and Grasp scores (5.00). Gemini 3.1 Flash Image ranks second ($\mu = 4.67$). Ministral 8B is last ($\mu = 3.91$) with the weakest Robustness (2.66) and highest latency (20.2s).

Table 2: Test 1 — updated mean subscores, final score with 95% bootstrap CI, and within-evaluator run standard deviation. Three runs per evaluator, two evaluators, 8 objects. Best per column in **bold**.

Model	A	B	C	D	E	Final	95% CI	$\bar{\sigma}_{\text{run}}$
GPT-5.4 Mini	4.86	5.00	5.00	5.00	4.53	4.88	[4.80, 4.94]	0.08
Gemini 3.1 Flash	4.87	4.94	5.00	4.80	3.75	4.67	[4.58, 4.75]	0.20
Seed 2.0 Lite	4.72	4.76	4.86	4.65	4.00	4.60	[4.16, 4.85]	0.39
Qwen 3.5 9B	4.03	4.84	4.83	4.59	3.12	4.28	[4.05, 4.52]	0.28
Ministral 8B	3.90	4.52	4.67	3.76	2.66	3.91	[3.68, 4.16]	0.50

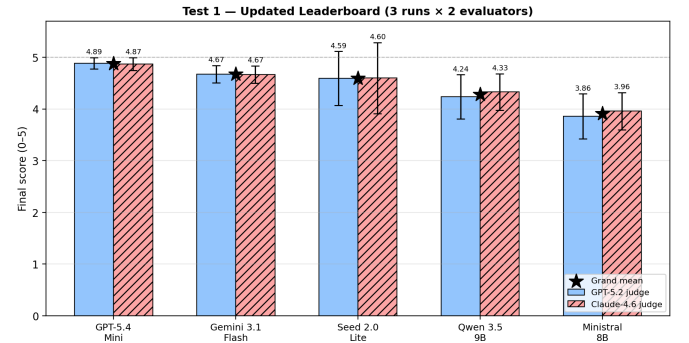


Figure 1: Test 1 — Updated leaderboard (3 runs $\times$ 2 evaluators). Bars: mean over 8 objects and 3 runs, per evaluator; error bars = across-object std. Stars: grand mean (both evaluators averaged).

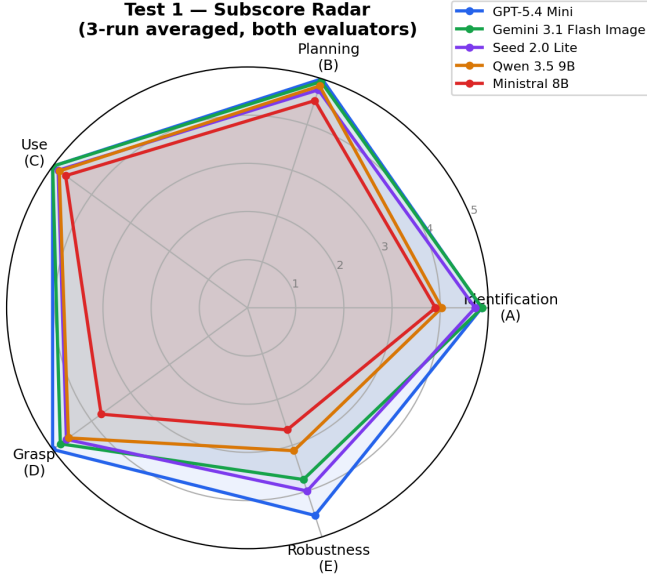


Figure 2: Test 1 — Subscore radar (3-run averaged, both evaluators).

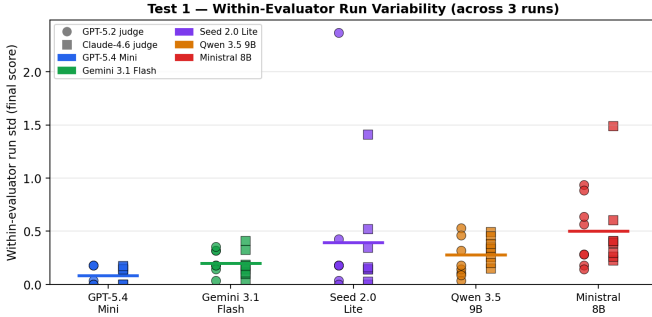


Figure 3: Test 1 — Within-evaluator run variability. Each point is the standard deviation of an evaluator’s three runs on one (model, object) cell. Horizontal bars show per-model means. GPT-5.4 Mini is the most stable ($\bar{\sigma} = 0.08$); Ministral 8B the most volatile ($\bar{\sigma} = 0.50$).

Ranking stability. The relative ordering is preserved (GPT-5.4 Mini > Gemini > Seed > Qwen > Ministral). Seed 2.0 Lite’s widened CI [4.16, 4.85] reveals real run-level instability ($\bar{\sigma}_{\text{run}} = 0.39$) partially masked in the original single-run estimate. Ministral 8B’s mean falls from 4.00 to 3.91, correcting an upward single-run bias.

Inter-evaluator agreement. With three runs per evaluator, agreement strengthens substantially: Pearson $r = 0.91$ (vs. 0.50 originally), Spearman $\rho = 0.89$, MAE = 0.16. The residual gap with Tests 2–3 ($r \approx 0.90$ –0.94) confirms that single-object open-ended evaluation remains inherently more subjective than structured localization tasks—but the earlier “moderate agreement” conclusion was largely an artifact of single-run noise.

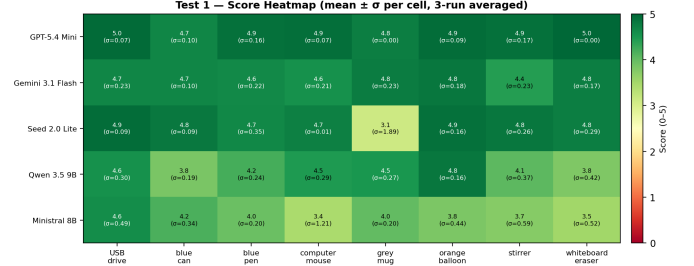


Figure 4: Test 1 — Score heatmap (mean / run σ per cell, 3-run averaged).

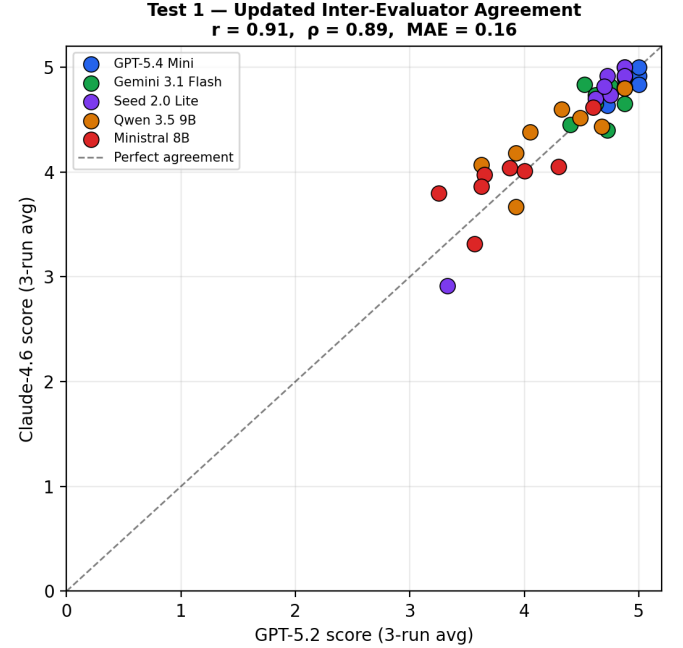


Figure 5: Test 1 — Updated inter-evaluator agreement on 3-run-averaged scores. Each point = one (model, object) cell ($n = 40$). $r = 0.91$, $\rho = 0.89$, MAE= 0.16.

Key observations: (i) Planning and Use are easy for all models (> 4.5); (ii) Robustness is the main differentiator; (iii) multi-run averaging substantially improves inter-evaluator reliability, demonstrating that single-run LLM-as-judge protocols systematically underestimate agreement in open-ended evaluation settings.

5 Test 2: Multi-Object Scene Grasping

5.1 Task

The VLM receives an image with *multiple* objects and a textual hint specifying the target (e.g., “blue pen”). The model must first *localize* the target in the scene (new `position_in_scene` field in the JSON) before reasoning

about grasp and use. The Identification criterion (A) thus evaluates localization accuracy rather than pure name recall.

5.2 Results

Table 3 reports Test 2 results. The top-2 ranking is preserved (GPT-5.4 Mini: $\mu = 4.80$; Seed 2.0 Lite: $\mu = 4.66$) while the bottom changes significantly.

Table 3: Test 2 — mean subscores and final score (both evaluators, 8 objects). Best per column in **bold**. * = includes one JSON failure (score 0).

Model	A	B	C	D	E	Final	RT
GPT-5.4 Mini	4.69	4.97	5.00	4.72	4.61	4.80	15.4
Seed 2.0 Lite	4.62	4.97	5.00	4.86	3.87	4.66	15.4
Gemini 3.1 Flash	4.59	4.83	4.89	4.75	3.79	4.57	17.4
Minstral 8B	2.56	4.73	4.81	4.53	2.58	3.84	17.9
Qwen 3.5 9B*	2.91	4.28	3.91	3.94	2.58	3.52	15.3

Localization failure in open-weight models. The most striking result is the collapse of Identification scores for Minstral 8B (A=2.56) and Qwen 3.5 9B (A=2.91). Both models struggle to reliably locate the target object within the scene, confirming that multi-object scenes pose a specific challenge for smaller open-weight VLMs.

JSON parse failure. Qwen 3.5 9B produces an invalid JSON response for the *stirrer* object across *both* evaluators (score = 0.00). If excluded from evaluation, Qwen 3.5 9B would achieve adjusted scores of A=3.33, B=4.89, C=4.47, D=4.50, E=2.95, and a final score of 4.02, which would have surpassed Gemini 3.1 Flash Image.

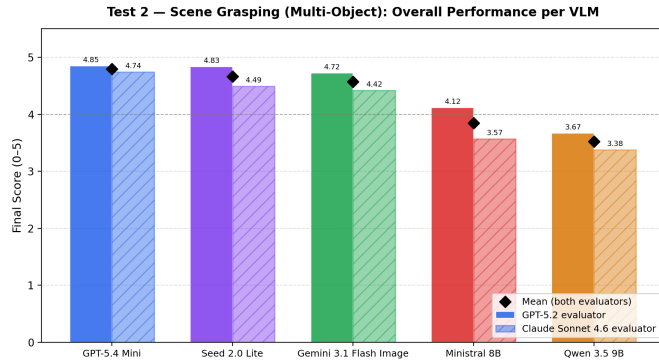


Figure 6: Test 2 — Final scores per VLM split by evaluator. Stars = mean.

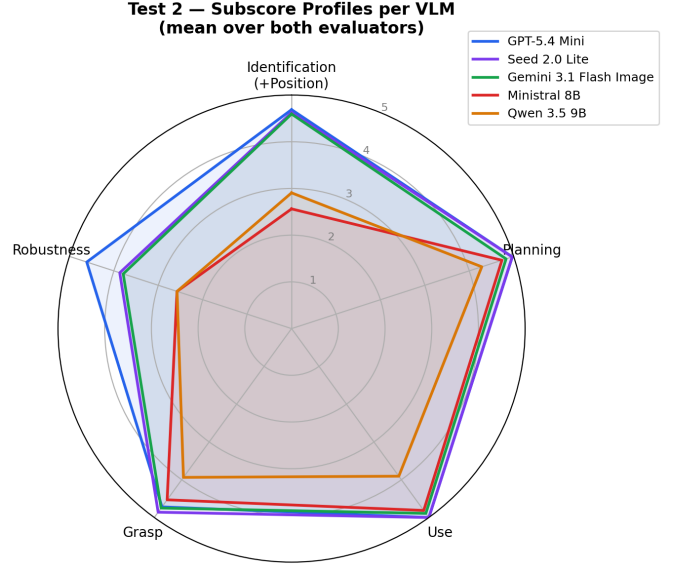


Figure 7: Test 2 — Subscore radar. Note the severe Identification and Robustness dip for Minstral 8B and Qwen 3.5 9B.

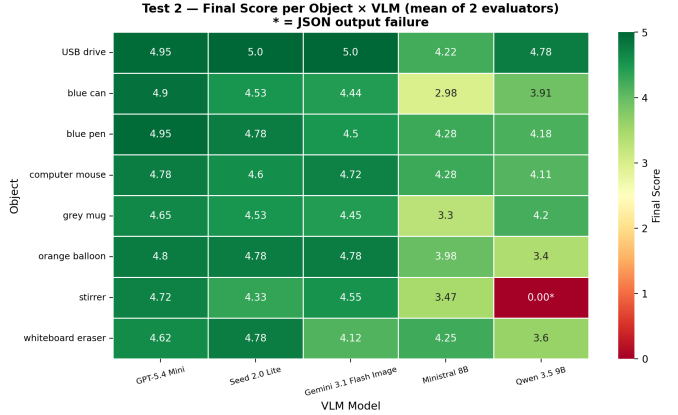


Figure 8: Test 2 — Score heatmap. * denotes the Qwen 3.5 9B JSON failure on *stirrer*.

Inter-evaluator agreement improves. Inter-evaluator agreement in Test 2 is *strong* ($r = 0.94$, MAE= 0.36). Clear localization failures produce unambiguous low scores that both LLM judges assign consistently.

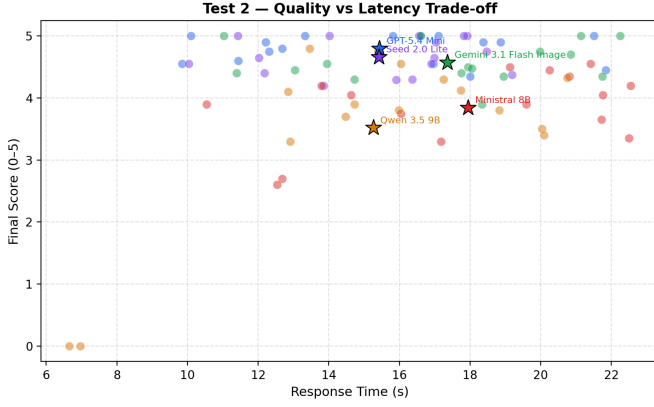


Figure 9: Test 2 — Quality vs latency. GPT-5.4 Mini and Seed 2.0 Lite jointly occupy the Pareto-optimal region.

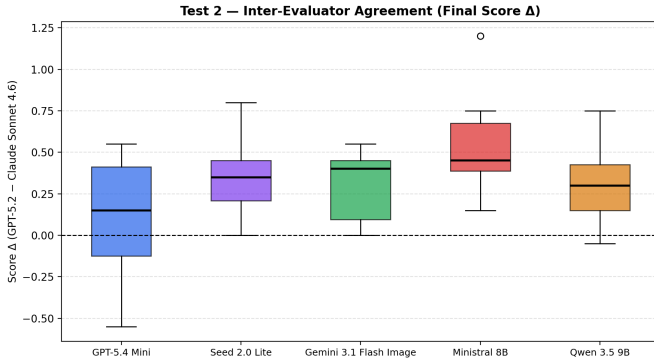


Figure 10: Test 2 — Inter-evaluator agreement. GPT-5.2 systematically scores higher than Claude Sonnet 4.6.

6 Test 3: Strict Localization Grasping

6.1 Task

Test 3 uses the same multi-object scene setup but tightens the Identification rubric for cluttered scenes. The `position_in_scene` field is now graded as *+1 per valid spatial descriptor* (orientation, where in the image, nearby objects) up to a maximum of 3 points. Describing the wrong object now yields a mandatory score of 0 for criterion A.

6.2 Results

Table 4 shows that the top two models remain stable while the open-weight models diverge sharply. No JSON failures were observed in Test 3.

Table 4: Test 3 — mean subscores and final score (both evaluators, 8 objects). Best per column in **bold**.

Model	A	B	C	D	E	Final	RT
GPT-5.4 Mini	4.66	4.95	5.00	5.00	4.06	4.73	14.7
Seed 2.0 Lite	4.56	4.88	5.00	4.84	3.75	4.61	15.8
Gemini 3.1 Flash	3.88	4.75	4.86	4.78	3.36	4.33	17.9
Qwen 3.5 9B	2.91	4.59	4.81	4.66	2.73	3.93	18.0
Ministral 8B	1.28	4.63	4.75	3.91	2.11	3.33	17.9

Ministral 8B collapses on localization. Ministral 8B’s Identification score of 1.28/5 is the lowest recorded across all three tests. Its position sub-score averages just 0.69/3.

Identification breakdown. The gap between proprietary and open-weight models is almost entirely driven by Position: GPT-5.4 Mini achieves 2.88/3 and Seed 2.0 Lite 2.75/3, while Ministral 8B scores only 0.69/3. Attribute scoring is more uniform (range 0.59–1.81).

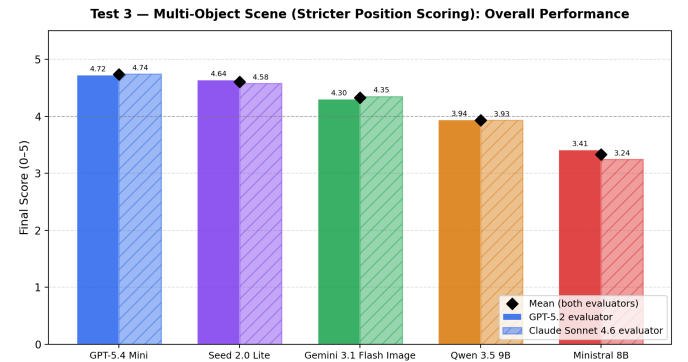


Figure 11: Test 3 — Final scores per VLM split by evaluator. Stars = mean.

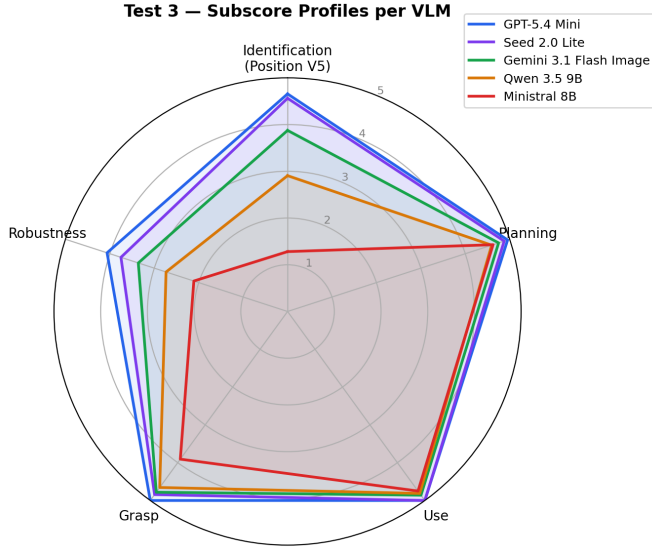


Figure 12: Test 3 — Subscore radar. Ministral 8B’s Identification spike inward reveals its localization failure.

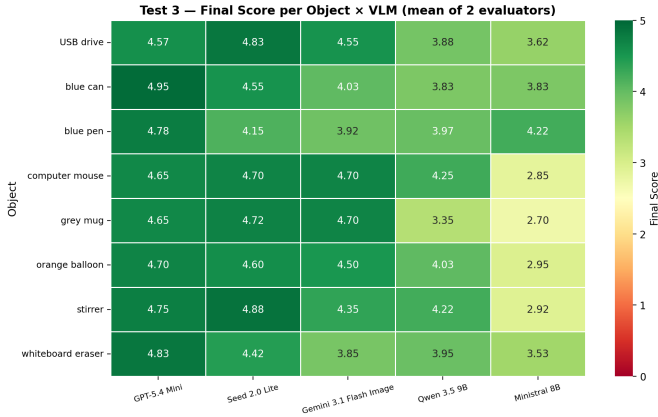


Figure 13: Test 3 — Score heatmap per object × VLM.

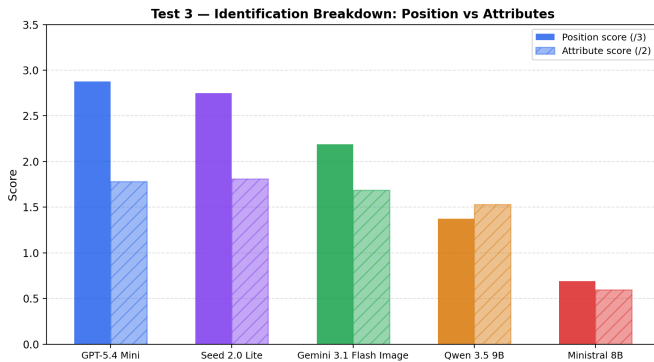


Figure 14: Test 3 — Identification decomposed into Position (solid) and Attribute (hatched) sub-scores.

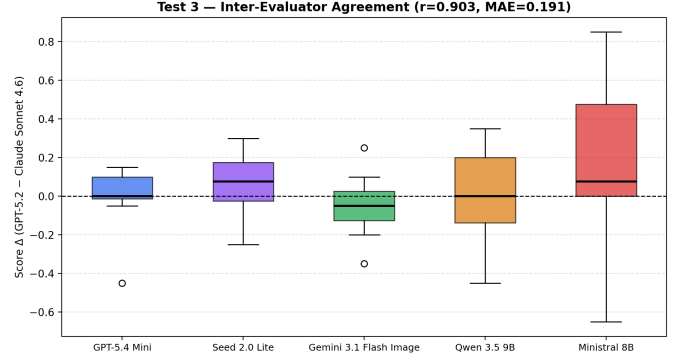


Figure 15: Test 3 — Inter-evaluator agreement ($r = 0.90$, $MAE = 0.19$).

	Test 1 (3 runs)	Test 2	Test 3
Pearson r	0.91	0.94	0.90
Spearman ρ	0.89	—	—
MAE	0.16	0.36	0.19

Figure 16: Inter-evaluator agreement across Tests 1–3.

7 Grasp Taxonomy Consistency Across Models

7.1 Motivation

Criterion D (Grasp) rewards correct grasp type selection—but this presupposes that both the evaluated VLM and the LLM judge share the same definition of each grasp type. To test this assumption, we probed all five models with a dedicated prompt asking them to define 10 standard grasp types (power, hook, cylindrical, tripod, precision, pinch, pincer, lateral, spherical, palmar) and characterise their contact geometry, force classification, and canonical objects. Definitions were compared against the established taxonomies of Napier [12], Cutkosky [13], and Feix et al. [14].

7.2 Convergence Summary

Table 5 summarises the degree of inter-model agreement for each grasp type. Grasps are colour-coded: **strong consensus** (4–5 models agree), **partial consensus** (2–3 models agree on key properties), **divergence** (fundamental disagreement on classification or contact geometry).

7.3 The Four Divergent Cases

1. Pinch grasp — contact surface conflict. All models agree that the pinch is a precision grasp opposing thumb and index, but diverge on the contact surface. GPT-5.4 Mini defines it broadly (pad *or* tip); Gemini and Qwen specify tip-to-tip exclusively; Seed uses pad-to-pad

Table 5: Inter-model grasp taxonomy consensus. $\checkmark$ = strong consensus, $\sim$ = partial consensus, $\times$ = significant divergence. “Key agreement” and “principal disagreement” are derived from pairwise comparison of model definitions against Napier–Cutkosky–Feix ground truth.

Grasp	Consensus	Point of agreement	Principal disagreement
Power grasp	$\checkmark$ Strong	Superclass, palmar contact, distributed force	Minor: exact role of thumb
Hook grasp	$\checkmark$ Strong	No thumb, IP flexion, traction load	Minor: degree of palmar contact
Cylindrical	$\checkmark$ Strong	Power, fingers in flexion, thumb opposed	No major disagreement
Tripod	$\checkmark$ Strong	Precision, 3 contact points, thumb+index+middle	Minor: Qwen misclassifies as “holding a cup”
Precision	$\checkmark$ Majority	Superclass, no primary palmar contact	Gemini too restrictive (“zero palm contact”)
Spherical	$\checkmark$ Majority	Power, fingers abducted	Seed calls it “hybrid”; others say power
Lateral grasp	$\sim$ Partial	Thumb + side of index	GPT & Ministral: precision; Gemini, Qwen, Seed: hybrid
Pinch grasp	$\sim$ Partial	Precision, thumb opposed to index	Pad-to-pad vs tip-to-tip: opposing definitions
Pincer grasp	$\times$ Divergent	Precision, very small objects	Three incompatible definitions (see Section 7.3)
Palmar grasp	$\times$ Divergent	Involves the palm	Precision (Gemini) vs Power (all others)

(palmar pinch, middle phalanges); Ministral uses pad-to-pad but for a different phalanx. Seed is the only model to explicitly distinguish pad-to-pad (pinch) from tip-to-tip (pincer), a distinction grounded in the Cutkosky taxonomy.

2. Pincer grasp — three incompatible definitions.

GPT-5.4 Mini defines pincer as tip-to-tip at two points (sub-type of precision); Gemini introduces a dynamic wrist-rotation component after contact (not supported by Napier or Feix); Qwen attributes a shear-dominant force profile (incorrect for a static precision grasp); Seed and Ministral converge on strict tip-to-tip. Gemini’s dynamic definition and Qwen’s force profile are particularly problematic: they would generate different motion primitives for the same word.

3. Palmar grasp — categorical inversion. Four models (GPT-5.4 Mini, Qwen, Seed, Ministral) classify the palmar grasp as a *power* grasp with dominant palmar contact. Gemini alone classifies it as *precision* (citing Feix ID 13), arguing that the palm is not the primary contact surface. While Feix’s classification is technically defensible in a narrow reading, Gemini’s definition creates an internal contradiction: calling a grasp “palmar” while specifying that the palm is absent from the primary contact.

4. Lateral grasp — classification boundary. GPT-5.4 Mini and Ministral classify the lateral grasp as precision (following Feix and Napier); Gemini, Qwen, and Seed

classify it as hybrid or mixed. The disagreement is legitimate: lateral grasps operate between precision and power regimes depending on the load, and the literature itself is divided.

7.4 Implications for Grasp Scoring (Criterion D)

These definitional divergences have a direct effect on benchmark validity. When a VLM generates “pinch grasp” for a small cylindrical object and a GPT-based judge evaluates it, both may use the term but mean different contact geometries—inflating the D score artificially. Conversely, Gemini’s unusual definition of “palmar” would cause it to be penalised by the GPT judge for a response that is internally consistent.

Table 6 shows the Grasp criterion (D) scores per model across Tests 1–3. The notably lower D scores for Ministral 8B (3.76, 4.53, 3.91) may partly reflect genuine grasp errors but may also reflect definition mismatches with the LLM judge on boundary grasps. This ambiguity motivates the inclusion of explicit grasp taxonomy references in future evaluation prompts.

Table 6: Grasp criterion (D/5) per model across Tests 1–3. Lower scores for open-weight models may reflect both genuine grasp errors and definition mismatches with the LLM judge.

Model	T1 (D)	T2 (D)	T3 (D)
GPT-5.4 Mini	5.00	4.72	5.00
Seed 2.0 Lite	4.65	4.86	4.84
Gemini 3.1 Flash	4.80	4.75	4.78
Qwen 3.5 9B	4.59	3.94	4.66
Ministral 8B	3.76	4.53	3.91

7.5 Visualisation of Convergence

The degree of inter-model consensus varies systematically with grasp complexity. Simple, archetypal grasps (power, hook, cylindrical) achieve near-universal agreement. Boundary grasps (lateral, pinch) show partial consensus. Grasps that sit at taxonomic interfaces (pincer, palmar) are genuinely divergent.

Grasp	Consensus
Power	100% 
Hook	95% 
Cylindrical	95% 
Tripod	90% 
Precision	85% 
Spherical	80% 
Lateral	55% 
Pinch	50% 
Pincer	40% 
Palmar	30% 

8 Test 4: Multi-Step Execution Planning

8.1 Task and Protocol Differences

Test 4 represents a fundamental shift in the evaluation paradigm. Each model receives: (i) a scene photograph, (ii) an ordered list of 5 manipulation tasks, and must output a valid JSON plan covering scene understanding, step-by-step object selection, grasp strategy, action type, pre/post conditions, and state transitions. Scores are on a $[0, 1]$ scale; 10 distinct real-world scenes are used; each (model, scene) pair is evaluated over **3 independent runs**; 7 scoring criteria are applied.

8.2 Scoring Protocol

OV (Object Validity), **GD** (Grasp Definition), **PPC** (Physical Properties), **AE** (Action Execution), **ST** (State Transition), **SU** (Scene Understanding), **PC** (Plan Consistency).

$$\text{Final} = \frac{\overline{OV} + \overline{GD} + \overline{PPC} + \overline{AE} + \overline{ST} + \overline{SU} + \overline{PC}}{7}$$

8.3 Results

Table 7: Test 4 — mean subscores and final score (scores $\in [0, 1]$). σ = run std. Best per column in **bold**.

	t								
Model	OV	GD	PPC	AE	ST	SU	PC	Final	σ
GPT-5	0.753	0.777	0.861	0.801	0.743	0.538	0.737	0.745	0.116
Gemini	0.766	0.798	0.888	0.822	0.707	0.522	0.772	0.751	0.107
Seed	0.733	0.773	0.885	0.805	0.630	0.564	0.668	0.722	0.095
Ministral	0.585	0.618	0.757	0.649	0.530	0.230	0.550	0.558	0.137
Qwen	0.429	0.444	0.509	0.469	0.410	0.293	0.414	0.425	0.391

Gemini Flash Image surpasses GPT-5.4 Mini. Gemini Flash Image takes the top position ($\mu = 0.751$) ahead of GPT-5.4 Mini ($\mu = 0.745$), leading on OV, GD, and AE. GPT-5.4 Mini compensates with a stronger State Transition score (0.743 vs 0.707).

Scene Understanding is the universal bottleneck. SU scores are low for all models (range 0.23–0.56).

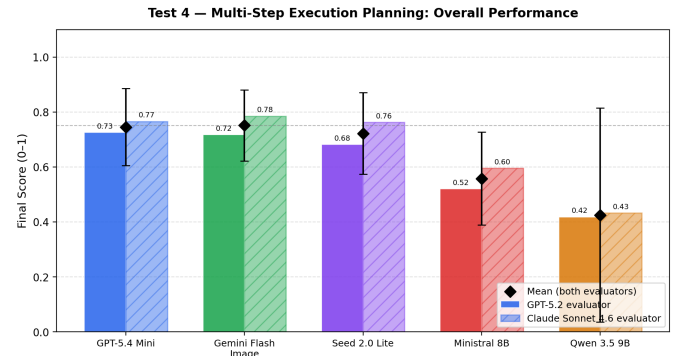


Figure 17: Test 4 — Final scores per VLM (error bars = run std). Gemini Flash Image outperforms GPT-5.4 Mini for the first time.

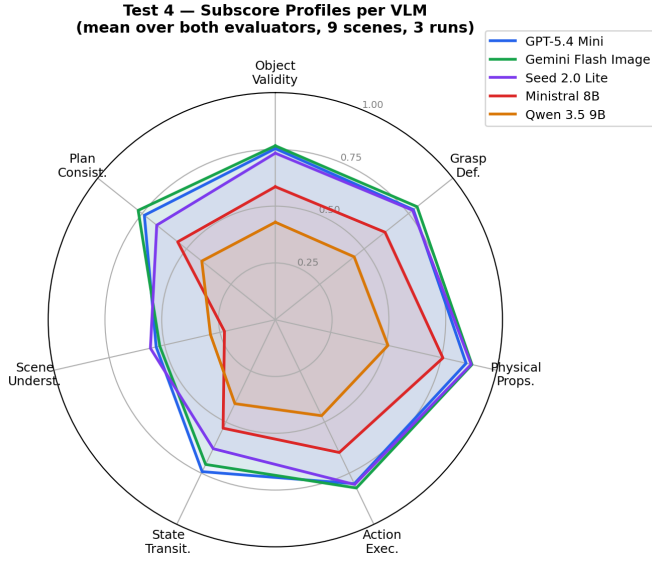


Figure 18: Test 4 — Subscore radar. All models dip on Scene Understanding.

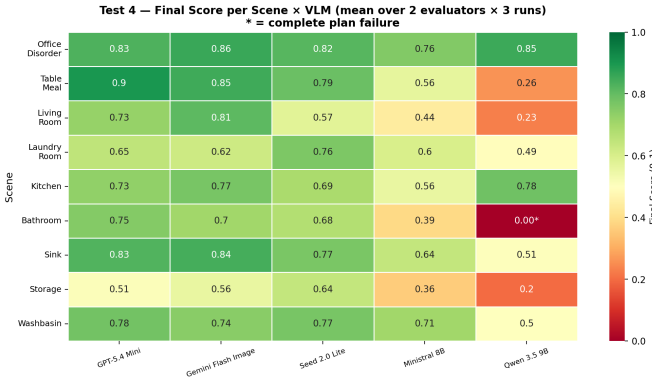


Figure 19: Test 4 — Score heatmap per scene × VLM.

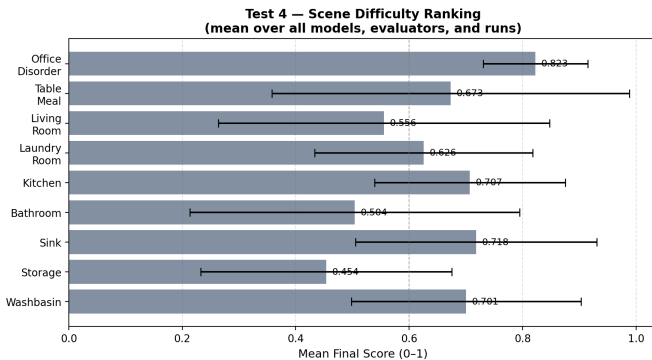


Figure 20: Test 4 — Scene difficulty ranking.

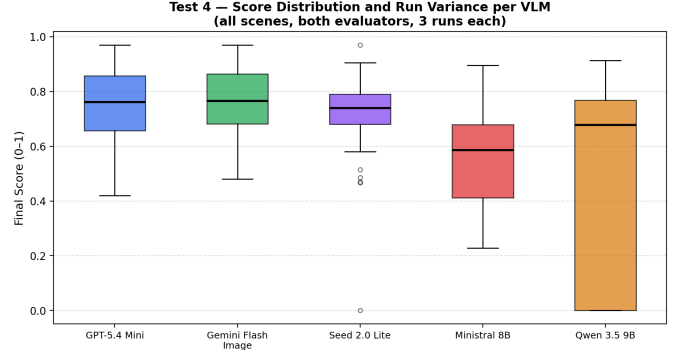


Figure 21: Test 4 — Score distribution per VLM.

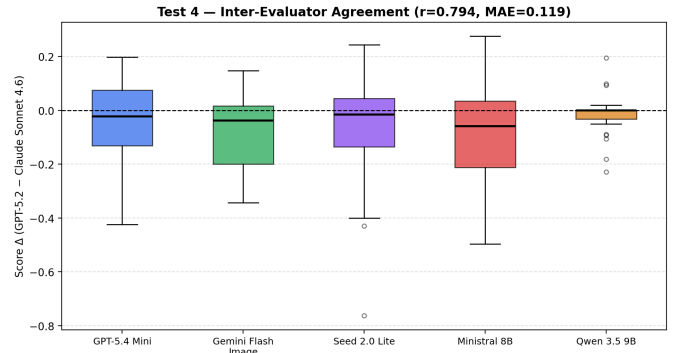


Figure 22: Test 4 — Inter-evaluator agreement ($r = 0.794$, $MAE = 0.119$).

9 Test 5: High-Level Task Planning

9.1 Task and Protocol Differences

Test 5 gives models only a **single high-level goal description** and requires them to autonomously decompose it into atomic sub-tasks via HTN decomposition (Phase 1) before generating a grounded execution plan (Phase 2). Each of the 10 base scenes is tested with 5 ambiguity variants (level 1 = detailed → level 5 = maximally vague), yielding 50 scene-task combinations evaluated over 3 runs (750 evaluations per judge).

9.2 Scoring Protocol

SU (Scene Understanding), **TD** (Task Decomposition — atomicity, grounding, no abstract verbs), **AP** (Action Plan, 11 sub-criteria), **GC** (Global Consistency). Scores on $\{0, 0.5, 1\}$.

$$\text{Final} = \frac{SU + \overline{TD} + \overline{AP} + GC}{4}$$

9.3 Results

Table 8: Test 5 — mean subscores and final score. Best per column in **bold**.

Model	SU	TD	AP	GC	Final	σ
GPT-5.4 Mini	0.699	0.805	0.816	0.830	0.788	0.094
Gemini Flash Image	0.685	0.817	0.828	0.810	0.785	0.086
Seed 2.0 Lite	0.634	0.769	0.758	0.749	0.726	0.202
Ministral 8B	0.429	0.659	0.618	0.626	0.578	0.147
Qwen 3.5 9B	0.390	0.503	0.493	0.494	0.464	0.386

GPT-5.4 Mini and Gemini Flash Image at parity. GPT-5.4 Mini leads on SU and GC; Gemini leads on TD and AP—complementary profiles suggesting different strengths.

Task Ambiguity Analysis. Performance increases with task ambiguity (level 1 $\mu = 0.652 \rightarrow$ level 5 $\mu = 0.714$). Detailed descriptions impose strict coverage requirements, while vague instructions allow models to select sub-tasks they are most confident about.

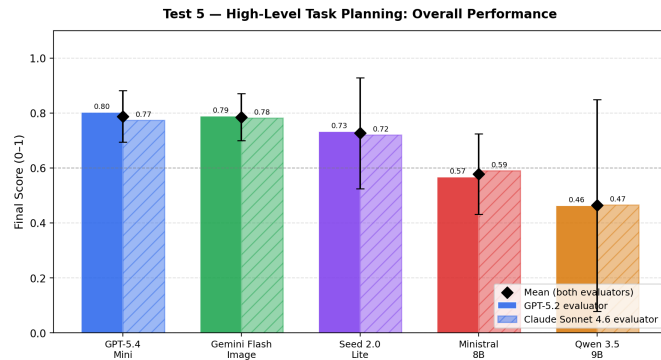


Figure 23: Test 5 — Final scores per VLM.

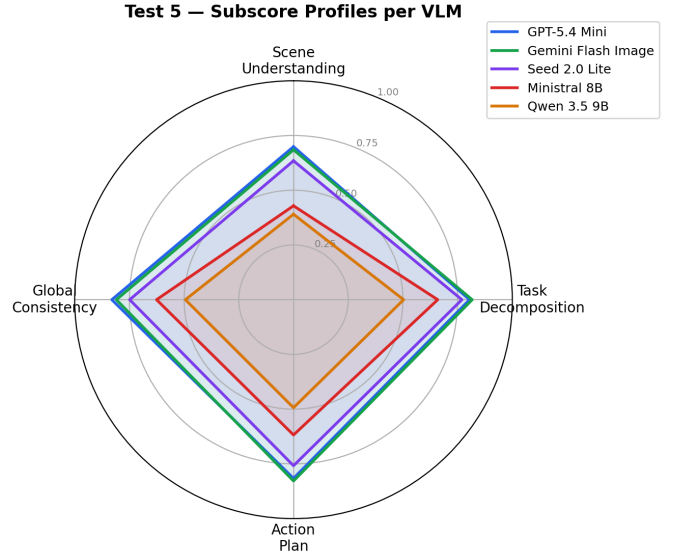


Figure 24: Test 5 — Subscore profiles.

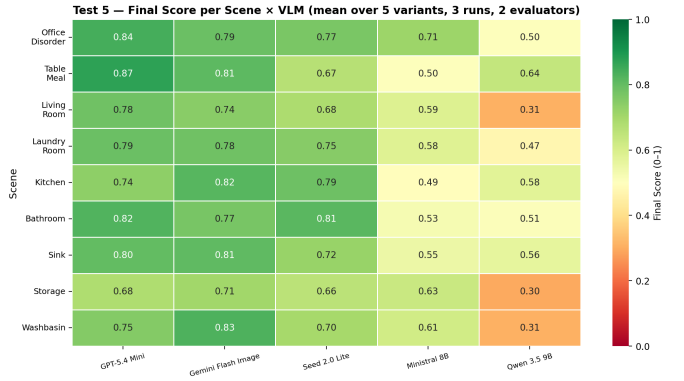


Figure 25: Test 5 — Score heatmap per base scene $\times$ VLM.

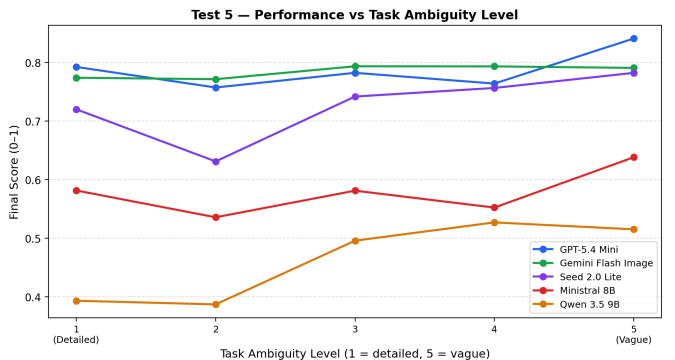


Figure 26: Test 5 — Score vs task ambiguity level.

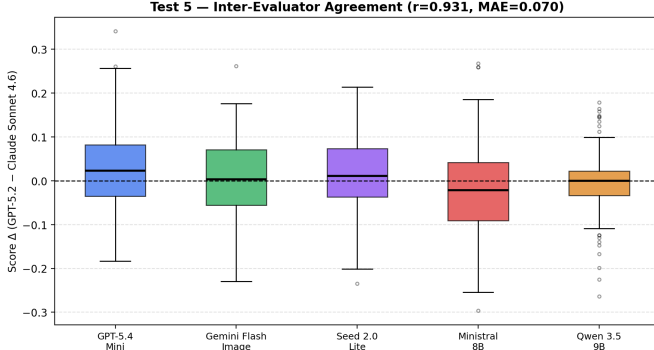


Figure 27: Test 5 — Inter-evaluator agreement ($r = 0.931$, $\text{MAE} = 0.070$).

10 Cross-Test Synthesis

Table 9: Global benchmark: normalized final scores (Tests 1–3 divided by 5; Tests 4–5 native $[0, 1]$). T1 updated with 3-run averaging.

Model	T1	T2	T3	T4	T5	Mean
GPT-5.4 Mini	0.976	0.960	0.947	0.745	0.788	0.883
Gemini Flash Image	0.934	0.914	0.866	0.751	0.785	0.850
Seed 2.0 Lite	0.920	0.932	0.922	0.722	0.726	0.844
Ministral 8B	0.782	0.768	0.666	0.558	0.578	0.670
Qwen 3.5 9B	0.856	0.704	0.786	0.425	0.464	0.647

(1) **GPT-5.4 Mini leads overall** (mean 0.883) with the lowest within-evaluator run variance ($\bar{\sigma} = 0.08$). (2) **Gemini’s planning strength**: joint-best in Tests 4–5 despite ranking 3rd in Tests 2–3. (3) **Seed 2.0 Lite: the stable all-rounder**, never below 3rd. (4) **The open-weight collapse**: Qwen 3.5 9B and Ministral 8B fall below 0.58 by Test 5.

11 Discussion

Task format drives model ranking. Gemini Flash Image ranks 2nd in Test 1, 3rd in Tests 2–3, and 1st in Test 4. No single test is sufficient to characterise a VLM for robotics.

Scene Understanding as the universal bottleneck. SU is the weakest criterion in Tests 4–5 for all models. Even the top models fail on roughly 30% of scene-level enumeration tasks.

Grasp definitions as a hidden evaluation risk. The taxonomy analysis reveals that VLMs may produce identical grasp labels for physically different contact configurations. This creates a hidden source of noise in criterion D scores, since a mismatch between the model’s definition

and the judge’s definition can inflate or deflate scores independently of actual grasp quality. Future benchmark designs should include explicit taxonomy anchors (e.g., citing Feix IDs) in both the generation and evaluation prompts.

Evaluator agreement improves with rubric structure. $r = 0.91$ (Test 1, 3 runs) $\rightarrow r = 0.931$ (Test 5). Single-run protocols underestimate true agreement by up to $\Delta r \approx 0.41$.

Limitations. Tests 1–3 use 8 controlled objects; Tests 4–5 cover 10 domestic scenes. GPT-5.2 consistently scores higher than Claude Sonnet 4.6 (leniency bias), and GPT-5.2 evaluating GPT-5.4 Mini introduces potential self-preference bias. Domestic scenes are Western-centric and may advantage models trained on Western-dominant data.

12 Conclusion

We presented a five-test benchmark scaling from single-object perception to autonomous hierarchical planning. The benchmark exposes a strict capability hierarchy: proprietary VLMs reach near-ceiling on perception but lose 15–25% under planning load, while open-weight VLMs collapse on spatial grounding.

A novel finding is that VLMs do not share consistent definitions of standard grasp types: strong consensus on simple grasps (power, hook) coexists with fundamental disagreement on boundary cases (pincer, palmar), a risk factor for robot systems that delegate grasp selection to VLMs without explicit taxonomy grounding.

Multi-run LLM-as-judge protocols are essential: single-run agreement underestimates true inter-judge reliability by up to $\Delta r \approx 0.41$.

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